

# Quantum Imaging Technical Documentation

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## Executive Summary

This document provides a comprehensive technical overview of quantum imaging principles, mathematical foundations, and practical implementation considerations. The content is designed to support verification and validation of quantum imaging systems.

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## 1. Fundamental Principles

### 1.1 Quantum State Representation

The quantum state of a photon is described by the wave function:

$$|\psi\rangle = \alpha|H\rangle + \beta|V\rangle$$

where  $|H\rangle$  and  $|V\rangle$  represent horizontal and vertical polarization states, and the normalization condition requires:

$$|\alpha|^2 + |\beta|^2 = 1$$

### 1.2 Entangled Photon Pairs

For entangled photon pairs in a Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|H\rangle_A|H\rangle_B + |V\rangle_A|V\rangle_B)$$

This represents maximum quantum correlation between photons A and B.

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## 2. Quantum Imaging Modalities

### 2.1 Ghost Imaging

Ghost imaging exploits quantum correlations to reconstruct images using photons that never interact with the object.

**Correlation Function:**

$$G^{(2)}(\mathbf{r}_1, \mathbf{r}_2) = \langle I(\mathbf{r}_1)I(\mathbf{r}_2) \rangle$$

where  $I(\mathbf{r})$  is the intensity at position  $\mathbf{r}$ .

**Image Reconstruction:**

$$I_{\text{ghost}}(\mathbf{r}) = \int G^{(2)}(\mathbf{r}, \mathbf{r}')T(\mathbf{r}')d\mathbf{r}'$$

where  $T(\mathbf{r}')$  is the object transmission function.

### 2.2 Quantum Illumination

The signal-to-noise ratio advantage in quantum illumination is given by:

$$\text{SNR}_{\text{quantum}} = \frac{4\eta^2 N_S^2}{N_B(N_B + 2\eta N_S)}$$

where  $\eta$  is the detection efficiency,  $N_S$  is the signal photon number, and  $N_B$  is the background noise photon number.

For comparison, classical illumination yields:

$$\text{SNR}_{\text{classical}} = \frac{\eta N_S}{N_B}$$

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## 3. Spontaneous Parametric Down-Conversion (SPDC)

### 3.1 Phase Matching Condition

Energy and momentum conservation require:

$$\omega_p = \omega_s + \omega_i$$

$$\mathbf{k}_p = \mathbf{k}_s + \mathbf{k}_i$$

where subscripts  $p$ ,  $s$ , and  $i$  denote pump, signal, and idler photons respectively.

### 3.2 Type-II SPDC Output State

$$|\psi\rangle_{\text{SPDC}} = \sum_{k_s, k_i} f(k_s, k_i) |1\rangle_{k_s, H} |1\rangle_{k_i, V}$$

where  $f(k_s, k_i)$  is the phase-matching function determining the spectral correlations.

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## 4. Quantum Coherence and Visibility

### 4.1 Visibility Definition

Interferometric visibility is defined as:

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

### 4.2 Coherence Time and Length

The coherence time relates to spectral bandwidth:

$$\tau_c = \frac{1}{\Delta\nu}$$

The coherence length is:

$$l_c = c\tau_c = \frac{c}{\Delta\nu} = \frac{\lambda^2}{\Delta\lambda}$$

where  $c$  is the speed of light,  $\lambda$  is the central wavelength, and  $\Delta\lambda$  is the spectral bandwidth.

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## 5. Quantum Detection and Measurement

### 5.1 Photon Counting Statistics

For a coherent state  $|\alpha\rangle$ , the photon number distribution follows Poissonian statistics:

$$P(n) = \frac{|\alpha|^{2n} e^{-|\alpha|^2}}{n!}$$

with mean photon number  $\langle n \rangle = |\alpha|^2$ .

### 5.2 Quantum Efficiency

Detector quantum efficiency affects measurement outcomes:

$$\eta = \frac{N_{\text{detected}}}{N_{\text{incident}}}$$

The effective quantum state after lossy detection becomes:

$$\rho_{\text{out}} = \sum_{n=0}^{\infty} \binom{N}{n} \eta^n (1 - \eta)^{N-n} |n\rangle \langle n|$$

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## 6. Quantum Imaging Resolution

### 6.1 Rayleigh Criterion (Classical Limit)

$$\Delta\theta_{\text{Rayleigh}} = 1.22 \frac{\lambda}{D}$$

where  $D$  is the aperture diameter.

### 6.2 Quantum-Enhanced Resolution

Using quantum states such as NOON states:

$$|\psi\rangle_{\text{NOON}} = \frac{1}{\sqrt{2}} (|N\rangle_a |0\rangle_b + |0\rangle_a |N\rangle_b)$$

enables sub-Rayleigh resolution:

$$\Delta\theta_{\text{quantum}} = \frac{\lambda}{ND}$$

providing an  $N$ -fold enhancement.

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## 7. Practical Implementation Considerations

### 7.1 Optical Path Length Stability

For interference visibility, path length must be stable to:

$$\Delta L < \frac{\lambda}{10}$$

Typical stabilization requirements:  $\Delta L < 50$  nm for visible wavelengths.

### 7.2 Timing Resolution

Coincidence detection requires timing windows:

$$\Delta t < \tau_c$$

Typical values:  $\Delta t \sim 1 - 10$  ns for narrowband SPDC sources.

### 7.3 Dark Count Rate

Single-photon detectors must have dark count rates:

$$R_{\text{dark}} \ll R_{\text{signal}}$$

Typical specifications:  $R_{\text{dark}} < 100$  Hz for avalanche photodiodes.

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## 8. Signal Processing and Image Reconstruction

### 8.1 Correlation-Based Reconstruction

The reconstructed image intensity is computed via:

$$I(\mathbf{r}) = \frac{1}{N} \sum_{i=1}^N D_{\text{bucket}}^{(i)} I_{\text{ref}}^{(i)}(\mathbf{r})$$

where  $D_{\text{bucket}}^{(i)}$  is the bucket detector signal and  $I_{\text{ref}}^{(i)}(\mathbf{r})$  is the reference beam intensity pattern for the  $i$ -th measurement.

## 8.2 Signal-to-Noise Optimization

The optimal measurement number scales as:

$$N_{\text{opt}} \propto \frac{1}{\text{SNR}_{\text{target}}^2}$$


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## 9. Quantum Advantage Metrics

### 9.1 Heisenberg Limit

The ultimate precision bound for phase estimation:

$$\Delta\phi \geq \frac{1}{N}$$

compared to the shot-noise limit:

$$\Delta\phi_{\text{classical}} \geq \frac{1}{\sqrt{N}}$$

### 9.2 Fisher Information

The quantum Fisher information for parameter estimation:

$$Q = 4(\langle \partial_{\theta} \psi | \partial_{\theta} \psi \rangle - |\langle \psi | \partial_{\theta} \psi \rangle|^2)$$

The Cramér-Rao bound states:

$$\Delta\theta^2 \geq \frac{1}{M Q}$$

where  $M$  is the number of measurements.

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## 10. Error Sources and Mitigation

### 10.1 Decoherence Effects

Decoherence time for maintaining quantum correlations:

$$T_{\text{dec}} = \frac{1}{\gamma}$$

where  $\gamma$  is the decoherence rate. Experimental requirement:  $t_{\text{exp}} \ll T_{\text{dec}}$ .

### 10.2 Alignment Tolerance

Angular misalignment tolerance for entangled photon systems:

$$\Delta\theta_{\text{align}} < \frac{\lambda}{2\pi w_0}$$

where  $w_0$  is the beam waist.

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## 11. Performance Validation Checklist

### System Parameters to Verify:

- Photon pair generation rate:**  $R_{\text{pairs}} = \text{_____ pairs/s}$
- Coincidence-to-accidental ratio (CAR):**  $\text{CAR} = \frac{R_{\text{coinc}} - R_{\text{acc}}}{R_{\text{acc}}} > 10$
- Quantum visibility:**  $V_Q > 0.7$  for quantum behavior
- Detection efficiency:**  $\eta_{\text{det}} = \text{_____ \%}$
- Timing jitter:**  $\sigma_t < \text{_____ ns}$
- Spatial resolution:**  $\Delta x = \text{_____ } \mu\text{m}$
- Integration time per frame:**  $t_{\text{frame}} = \text{_____ ms}$

### Key Validation Tests:

- Bell inequality measurement:  $S > 2$  (CHSH inequality)

- Quantum state tomography: Fidelity  $F > 0.9$
  - Interferometric visibility scan
  - Background subtraction verification
  - Dynamic range assessment
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## 12. References and Further Reading

### Foundational Papers:

- Shih & Alley (1988): Two-photon entanglement in type-II parametric down-conversion
- Pittman et al. (1995): Optical imaging by means of two-photon quantum entanglement
- Lloyd (2008): Enhanced sensitivity of photodetection via quantum illumination

### Recommended Texts:

- Fox, M. *Quantum Optics: An Introduction* (Oxford University Press)
  - Gerry, C. & Knight, P. *Introductory Quantum Optics* (Cambridge University Press)
  - Walls, D. & Milburn, G. *Quantum Optics* (Springer)
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## Appendix: Notation Conventions

| Symbol         | Description                                            |
|----------------|--------------------------------------------------------|
| $ \psi\rangle$ | Quantum state (ket notation)                           |
| $\hbar$        | Reduced Planck constant ( $1.055 \times 10^{-34}$ J·s) |
| $\omega$       | Angular frequency (rad/s)                              |
| $\lambda$      | Wavelength (m)                                         |
| $\mathbf{k}$   | Wave vector ( $\text{m}^{-1}$ )                        |
| $\eta$         | Quantum efficiency (dimensionless)                     |
| $\rho$         | Density matrix                                         |



| Symbol                  | Description       |
|-------------------------|-------------------|
| $\langle \cdot \rangle$ | Expectation value |
|                         |                   |
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